AER1301: MOMENT CLOSURE METHODS FOR KINETIC EQUATIONS OF COMPLEX TRANSPORT PHENOMENA

Assignment #2

1. At a given instance in time, t, the number density function, $\mathcal{F}$, of a spatially uniform monatomic gas of molecular mass, m, is

$$\begin{aligned}\mathcal{F}(\mathbf{v},t) = \mathcal{F}(v_x, v_y, v_z, t) \;&=\; n_1 \left(\frac{m}{2\pi k T_1}\right)^{3/2} \exp\left\{-\frac{m}{2kT_1}\left[(v_x - u_{1x})^2 + (v_y - u_{1y})^2 + (v_z - u_{1z})^2\right]\right\} \\ &+\; n_2 \left(\frac{m}{2\pi k T_2}\right)^{3/2} \exp\left\{-\frac{m}{2kT_2}\left[(v_x - u_{2_x})^2 + \left(v_y - u_{2_y}\right)^2 + (v_z - u_{2_z})^2\right]\right\},\end{aligned}$$

   where $\vec{v} = [v_x, v_y, v_z]$ is the total particle velocity and n_1, n_2, $\vec{u}_1 = [u_{1_x}, u_{1_y}, u_{1_z}]$, $\vec{u}_2 = [u_{2_x}, u_{2_y}, u_{2_z}]$, T_1, and T_2 are scalar and vector parameters defining $\mathcal{F}$.

   (a) What is the density of the gas?

   (b) What is the average (bulk) velocity of the gaseous particles? Provide expressions for each component of the average velocity vector.

   (c) What is the gas pressure?

   (d) If the gas above is subsequently allowed to undergo a large number of interparticle collisions, provide an expression for the equilibrium distribution function.

   (e) What would be the corresponding density, average (bulk) velocity, and pressure for the gas under the equilibrium conditions?

   Please show how you calculate the preceding values when providing your answers.

2. The initial number distribution function for a spatially uniform (homogeneous) gas is given by $\mathcal{F}(\mathbf{v}, t=0) = \mathcal{F}_\circ(\mathbf{v})$.

   (a) Assuming that there are no external forces, provide the form of the Boltzmann equation with the Boltzmann collision operator represented by the relaxation-time (BGK) approximation (i.e., the so-called BGK kinetic equation) corresponding to this space homogenous case where $\mathcal{M}(\mathbf{v}, t)$ is the Maxwell-Boltzmann distribution to which F is relaxing and τ is the relaxation time.

   (b) Assuming that the relaxation time, τ, is constant, it can be shown that, in the case of space homogeneous flow,

$$\mathcal{F}(\mathbf{v},t) - \mathcal{M}(\mathbf{v},t) = \left(\mathcal{F}_\circ(\mathbf{v}) - \mathcal{M}_\circ(\mathbf{v})\right) e^{-\frac{t}{\tau}} = \left(\mathcal{F}_\circ - \mathcal{M}_\circ\right) e^{-\frac{t}{\tau}}\,.$$

   Using the preceding result, derive an expression for the distribution function $F(\mathbf{v}, t)$ as a function of time, t, which satisfies the BGK kinetic equation in the space homogenous case.

3. Briefly summarize the main similarities and differences between the Chapman-Enskog and Grad 20-moment approaches for obtaining non-equilibrium solutions of the Boltzmann equation. Be sure to include in your discussion the following items:

(a) a comparison of the approximate forms for the non-equilibrium number density function (NDF) used in each approach,

(b) a comparison of the macroscopic quantities (velocity moments) that are required to describe the distributions functions in each approximation,

(c) a comparison of the generalized transport equations that must be solved to prescribe the macroscopic velocities moment in each of the two approaches.

4. (a) The 13- and 20-moment closures were postulated by Grad for providing approximate solutions to the Boltzmann equation and thereby describing non-equilibrium rarefied flows. A simplification of the 13- and 20-moment models is provided by the 10-moment model. What velocity moments are involved in this simplified closure and show how the model can be obtained from the 20-moment closure (i.e., indicate what assumptions/approximations are used)? Why is the simplified model called a "10-moment" closure.

(b) Provide an expression for the approximate number density function (NDF), $\mathcal{F}^{(10)}$, of the 10-moment model.

(c) Assuming that there are no external forces acting on the particles, give the moment equations for the 10-moment model and indicate which terms are associated with collisional processes (i.e., the "collision" terms). Using the expression for the 10-moment distribution function found in part (b) and adopting a relaxation time or BGK approximation for the collision integral, show how these collision terms may be evaluated. Evaluate the appropriate collision terms and indicate which moments are collisional invariants.

(d) The 10-moment model is said to be deficient in the sense that it does not possess a physically correct Navier-Stokes (continuum flow) limit. Indicate why this is so. Does the 10-moment model possess a valid local thermal equilibrium (LTE) or Euler limit (indicate why or why not)?

(e) Assuming that the relaxation time for particle collisions, τ, is short, apply a Chapman-Enskog-like perturbative expansion technique to the fluid stress equations of the 10-moment model found in part (c) and derive a first-order approximation for the viscous stresses. Please provide all of the important steps in your calculation. Compare your results to those for the Navier-Stokes equations. What is the value of the dynamic viscosity that results from your calculation?